Team 26061

AQUABOTS Aeropak

System Requirements Document (SRD)

Revision: 3 (updated 11/18/2025)

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Revisions:

10/9/2025:

- Updated bottom range of ocean temperatures to 0 degrees Celsius, to reflect that drones will not be going into below freezing conditions.
- Waterproofing verification added.

10/30/2025:

- Added System Requirements for Drone-to-Drone Communications
- Added System Requirements for Swarm Control

11/18/2025:

- Removed/Modified All Requirements Regarding Deployment Mechanism
- Minimum of three drones needed to demonstrate swarm abilities
- Threshold and objective values added for battery life, speed, and internal volume.

1.0 Introduction and Scope

1.1 Introduction

This document outlines the system requirements for a swarm-based aquatic drone network being developed for the Arizona Center for Accelerated Biomedical Innovation (ACABI) at the University of Arizona. The current design phase focuses on enhancing swarm communication via a streamlined GUI for sensor and camera operation, and dual-mode operation allowing both autonomous and manual control.

Each aquatic bot will collect critical environmental data, including pH, salinity, and temperature measurements at a constant depth, to provide deeper insight into ocean health and ongoing climate change. The system will withstand oceanic conditions while maintaining self-righting capabilities and durable waterproofing. Swarm functionality will enable coordinated missions, increased spatial coverage, and improved data reliability. This project aims to deliver a resilient, adaptive, and cost-effective platform for distributed ocean health assessment and environmental research.

1.1 Project Scope

The AQUABOTS Aeropak is a remotely controlled, modular swarm of water-borne drones capable of surviving ocean conditions, self-righting, and performing multi-parameter environmental sensing. The system integrates propulsion, communication, sensor and power subsystems to enable long-duration autonomous data collection with minimal operator oversight.

Is	Is Not
A functional aquatic drone system remotely operated via satellite and network links	A fully autonomous system independent of human oversight
Designed for deployment into open-ocean, lake, or river environments	Designed for use on land or non-aquatic surfaces
Operates in water to collect temperature, salinity, pH, and visual data	Include undefined or unapproved sensors outside the project scope
Displays GPS position, sensor data, and video on a ground-station GUI	Provide in-depth onboard analytical or predictive software beyond mapping and telemetry
Powered by internal battery systems with optional solar augmentation	Depend on a tethered or constant external power source

Is	Is Not
Waterproof to IP68 and compliant with MIL-STD-810H environmental test methods	Intended for use in sea states greater than Category 3 or extreme weather events
Uses commercial off-the-shelf (COTS) hardware where feasible to maintain cost efficiency	Employ proprietary or export-restricted components

This scope defines a **bounded prototype system** intended to demonstrate self-righting stability, satellite-based telemetry, and basic swarm-network functionality. It does not include long-term unattended operation or deep-ocean missions.

1.2 Stakeholders

Stakeholder	Role / Interest
AQUABOTS Capstone Team	Design, integration, testing, and verification of all system subsystems.
Sponsors (E. Stocker, M. Slepian)	Provide technical, biomedical, and oceanographic guidance.
University of Arizona College of Engineering	Oversight, grading, and project compliance under ENGR 498 A/B.
Regulatory Authorities (FAA, USCG, IMO)	Ensure compliance with maritime navigation, and environmental standards.
End Users (Research Labs, Coast Guard, Environmental Agencies, E Stocker, M. Slepian)	Utilize collected data for ocean monitoring, pollution mapping, or infrastructure inspection.

2.0 Referenced Documents

2.1 Enclosure Protection and Environmental Durability

- **IEC 60529:2013 (IP Code)**: Degrees of protection for enclosures; target IP68 for immersion.
- **MIL-STD-810H (2019)**: Environmental tests (shock, vibration, temp, humidity, salt fog, immersion).
- **MIL-STD-461G (2015)**: EMI/EMC limits and test methods for electronic subsystems.

- **IEC 60945:2002+A1:2008**: Maritime nav/radio equipment; environmental and EMC requirements.
- **ISO 12215-2:2002 (confirmed 2023):** Small craft; Hull construction and scantlings; Design pressures and structural requirements for craft < 24 m; used for composite hull design verification and future commercial certification.

2.3 Marine Environmental Protection

- **IMO MARPOL Annex V**: Prohibits discharge of plastics/garbage at sea; recovery or biodegradable only.

2.4 Communication and Identification

- **ITU-R M.1371**: AIS technical characteristics and message protocols.
- **IEC 61993-2**: AIS Class A shipborne equipment; performance and testing.
- **IEC 62287-1/-2**: AIS Class B equipment; performance and testing (System will be AIS-ready but not equipped for current testing).
- **FCC Part 80 (US)**: Maritime services radio rules (licensing/operation for AIS/VHF).
- **FCC Part 15 (US)**: Unlicensed device emissions (Wi-Fi/LoRa/compute EMC).

2.5 Hull Construction and Stability

- **UNE-EN ISO 12217-3:2022**: Small craft; stability and buoyancy assessment (<6 m).
- **ASTM F3052-14(2020)e1**: Small boat stability testing; deadweight survey and air-inclining.
- **COLREGs (1972, as amended)**: International collision-avoidance rules; lights/shapes/behaviors for vessels.

2.6 Power, Battery, and Product Safety

- **UN/DOT 38.3**: Lithium cell/battery transport tests (air/sea/road shipment).
- **IEC 62133-2 / UL 1642**: Rechargeable lithium battery safety (design/pack certification).
- **UL 62368-1 / IEC 62368-1**: Safety requirements for low-voltage electronic and communication equipment; applied as design guidance for the prototype, required for commercial certification.
- **CE / UKCA Marking Framework:** Comprehensive conformity assessment combining radio, EMC, and safety directives for products sold in the European and UK markets. Exempt for prototype testing but required for final commercial release.

2.7 Sensors and Data Quality

- **ISO 22013:2021**: Marine environment sensors; performance specification, testing, reporting.

2.8 System Safety and Electromagnetic Effects (govt./military users)

- **MIL-STD-882E**: System safety; hazard analysis and risk mitigation process.

2.9 Documentation, Quality, and Conformity

- **ISO 9001:2015**: Quality management systems (process assurance; optional but useful).

3.0 Definitions, Acronyms, and Abbreviations

Term	Definition
AIS	Automatic Identification System, VHF maritime identification and tracking.
AWG	American Wire Gauge for conductor size (e.g., 8–12 AWG power, 24–26 AWG signal).
Cat6 / RJ45	Shielded Ethernet cable and connector used for Jetson↔Starlink.
COLREGs	International Regulations for Preventing Collisions at Sea.
COTS	Commercial off-the-shelf.
EMI	Electromagnetic interference.
ESC	Electronic Speed Controller for brushless motors.
GIS	Geographic Information System.
GUI	Graphical User Interface for ground station.
I²C	Two-wire serial bus for sensors/power monitors.
IMU	Inertial Measurement Unit.
IP67	Ingress rating used on external Starlink connectors.
IP68	Ingress Protection rating: dust-tight, immersion protection.

Jetson	NVIDIA Jetson Nano mission computer.
LoRa	Long Range low-power radio.
MJPEG	Motion-JPEG stream format used in GUI.
MQTT	Message Queuing Telemetry Transport protocol.
PV	Photovoltaic solar augmentation
PWM	1–2 ms, 50 Hz control to ESCs
QoS (Quality of Service)	MQTT or ROS 2 message delivery reliability level defining how guaranteed message delivery is (QoS 0: at most once, QoS 1: at least once, QoS 2: exactly once)
RJ45	Registered Jack-45 modular 8P8C Ethernet connector
ROS 2 / ROS 2 Foxy	Robot Operating System 2 middleware used on the Jetson
RTSP / MJPEG	Real-time streaming protocols/codec used for video
SRVM	System Requirements Verification Matrix
Starlink	Satellite internet terminal used as primary uplink
TLS	Transport Layer Security
UART	3.3 V serial link ESP32↔Jetson and sensors.
VHF	Very High Frequency radio band (30–300 MHz)
VM	Cloud virtual machine hosting broker/GUI.
EMC	Electromagnetic Compatibility

4.0 Overall System Description

4.1 System Overview

The AQUABOTS AeroPak system is an autonomous aquatic-surface drone designed to collect, process, and transmit environmental and visual data from marine and freshwater environments. Each drone integrates propulsion, navigation, power, sensing, communication, and data-processing subsystems within a rugged, IP68-rated composite enclosure. After release, the

vehicle self-rights to an upright position, activates its sensors, and establishes a satellite uplink to transmit telemetry and video to a remote ground station.

The system supports networked swarm operation through Wi-Fi mesh or LoRa backup links, enabling multiple units to share data and coordinate coverage areas. A single operator can monitor and control all deployed drones through a ground-station GUI connected to a secure cloud server.

4.2 Operational Context

The AQUABOT platform operates on the water surface in open oceans, coastal zones, lakes, and rivers under environmental conditions up to Sea State 3. Typical missions involve rapid deployment from shore, boats, or aircraft with addition of parachute and deployment mechanism. After splashdown, the drone performs autonomous stabilization and begins real-time data acquisition.

Sensors mounted on the underside of the hull remain submerged to measure temperature, salinity, pH, and conductivity. A waterproof camera provides continuous video streaming at 1080 p 30 fps. All collected data are transmitted via Starlink satellite uplink to the ground-station interface for visualization, storage, and analysis.

Power is provided by a 0.8 – 1.3 kWh battery pack with optional solar augmentation. The system is designed for at least 5 hours of mission endurance and can be recovered manually or by service vessel after operation. All components are compliant with MARPOL Annex V principles to prevent discharge of plastics or debris.

4.3 System Architecture

The system consists of the following major subsystems, interconnected through defined power and data interfaces:

Subsystem	Primary Functions	Key Interfaces
Power Subsystem	Energy storage, distribution, regulation	Battery → PDB → buck converters (5 V/10 A Jetson, 5 V/2 A ESP32)
Propulsion Subsystem	Dual ESC-controlled thrusters for maneuvering	PWM signals from ESP32; 12–24 V bus power
Flight Controller Subsystem	Low-level control, IMU/GPS fusion, failsafes	UART to Jetson; I²C to sensors

Mission Computer & Video Subsystem	High-level navigation, ROS 2 nodes, video encoding	UART to ESP32; Ethernet to Starlink
Satellite Uplink Subsystem	Broadband telemetry/video link via Starlink	Ethernet Cat6 RJ45; 12–24 V @ 2–3 A
Sensors & Sampling Subsystem	Environmental data collection	I²C/UART to ESP32; sensor array on underside
Self-Righting & Failsafes Subsystem	Automatic recovery and emergency stop logic	IMU trigger inputs; PWM cutoff outputs
Ground Station & Cloud Subsystem	Command, visualization, data storage	MQTT broker and web GUI via Starlink network

4.4 Key Capabilities and Features

- Self-righting within 10 s of capsize or inversion.
- Real-time 1080 p video streaming and telemetry at ≥ 1 Hz via Starlink.
- Secure MQTT over TLS communication with cloud database and GUI.
- Optional LoRa mesh heartbeat backup when satellite link is lost.
- IP68 enclosure and MIL-STD-810H environmental durability design.
- Five-hour operational endurance with solar charging offset of 30 – 80 W.
- Modular architecture for swarm deployment and multi-unit coordination.

4.5 Interfaces and Dependencies

The AQUABOT system relies on defined internal and external interfaces:

4.5.1 Internal Interfaces

- **Power:** PDB distributes regulated voltage rails to ESCs, Jetson, ESP32, and Starlink.
- **Data:** UART links connect ESP32 with Jetson; Ethernet connects Jetson to Starlink; I²C links support sensor buses.
- **Control:** PWM lines drive ESCs; IMU and tilt sensors feed self-righting logic.

4.5.2 External Interfaces

- **Satellite Network:** Starlink link for data uplink/downlink (IPv4/TCP over MQTT and RTSP).
- **Ground Station and Cloud:** Operator GUI and database for command and telemetry processing.

- **Environmental Medium:** Operates at air–water interface; sensors submerged ≤ 0.3 m.

5.0 Requirements

5.1 Performance Requirements

- **5.1.2 Swarm Control:** All AQUABOT drones shall autonomously self-right and enter operational state within 60s of water impact.
- **5.1.3 Stability:** Each aquatic AQUABOT drone shall self-right and maintain operational orientation IAW UNE-EN ISO 12217-3:2022 and/or ASTM F3052-14(2020)e1.
- **5.1.4 Speed:** Each aquatic AQUABOT drone shall be able to sustain a speed of 2.5 kts in calm water conditions.
- **5.1.5 Battery Life:** Each AQUABOT drone shall have a minimum threshold operational runtime of 5 minutes at nominal load (motors + Starlink + camera streaming), with an objective operational runtime of 5 hours.
- **5.1.6 Data Collection:** Each AQUABOT drone shall measure temperature, salinity, pH and capture visual imagery at a minimum of 1080p resolution, 30 fps.
- **5.1.7 Data Mapping:** Each AQUABOT drone shall generate a georeferenced map of collected environmental variables and export data in GIS-compatible format.

5.2 Design Requirements

- **5.2.1 Hull Shape:** Each AQUABOT drone shall have a hydrodynamic hull shape generating a drag coefficient ≤ 0.5
- **5.2.2 Internal Volume:** Each main AQUABOT drone shall provide a minimum usable internal volume of 0.07 m^3 for electronics, power systems, and sensors. Subunits will have an internal volume of 0.001 m^3, since space for the Starlink antenna is not needed.
- **5.2.3 Materials:** The system shall be composed of corrosion-resistant materials suitable for prolonged saltwater exposure..
- **5.2.4 Payload Mounts:** The hull shall provide standardized, sealed mounts for Starlink terminal, battery, and sensor payload.

5.3 Environmental Requirements

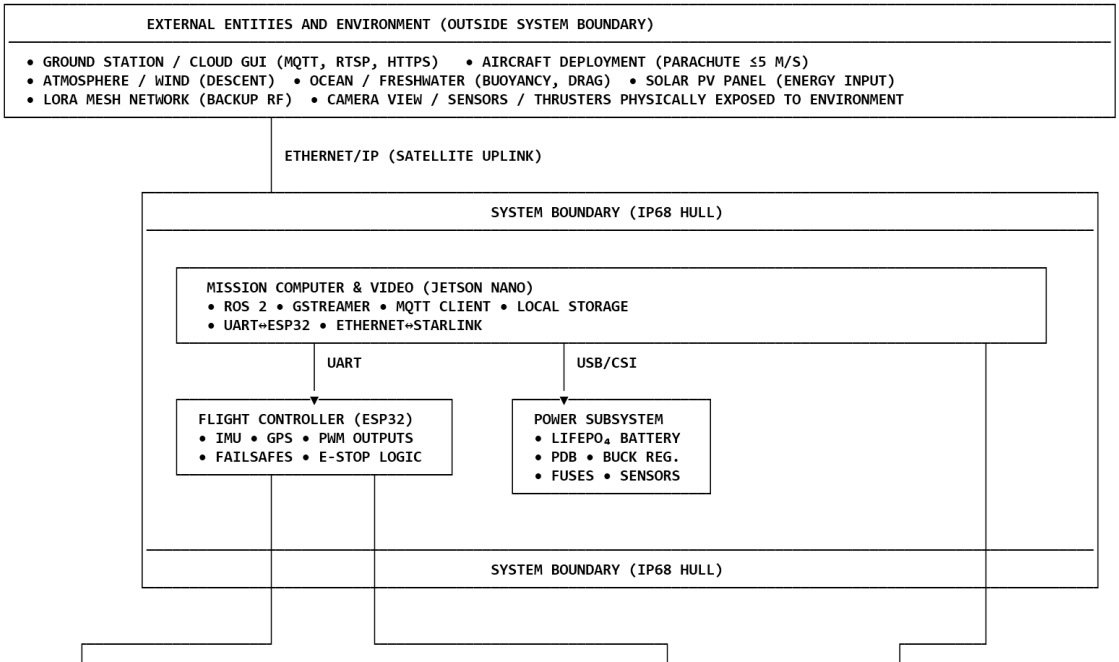
- **5.3.1 Survivability:** The system shall be designed to survive and operate in up to Sea State 3 conditions IAW ISO 12215-2:2002.

- **5.3.2 Waterproofing:** Each aquatic AQUABOT drone shall meet IP68 ingress protection standards IAW IEC 60529:1989 + AMD1:1999 CSV. Waterproof components shall be tested in 1.5 m depths to ensure compliance.
- **5.3.3 Operating Temperature:** Each AQUABOT drone shall operate in ambient water temperatures between –10 °C and +35 °C.
- **5.3.4 Non-Hazardous Surfaces:** The system shall not contain harmful external surfaces that could damage personnel, equipment, or marine life.

5.4 Interface Requirements

- **5.4.1 Autonomy & Control:** Each AQUABOT drone shall support both autonomous navigation and remote operator control via a satellite-based waypoint system.
- **5.4.2 Inter-Drone Communication:** Each AQUABOT drone shall support peer-to-peer communication within 100m using the ESP-NOW protocol implemented on its ESP32 microcontroller.
- **5.4.3 Data Relay** AQUABOT drones shall transmit collected sensor and positional data to the command unit which shall serve as the primary relay node.
- **5.4.4 Satellite Uplink:** Each AQUABOT drone shall independently transmit live and stored data to a remote ground station via its onboard Starlink terminal.
- **5.4.5 Display:** The display shall show half the display with Geo-overlay and the other half will display live-streaming.
- **5.4.6 Swarm Control:** The AQUABOT swarm shall maintain coordinated network behavior through a leader-follower model executed via Jetson Mission Computer and MQTT topics. Each drone shall publish and subscribe to swarm state messages at ≥ 1 Hz with < 1 s latency.

5.5 System Block Diagram



6.0 Verification Requirements

Verification confirms that each AQUABOT system requirement is satisfied through planned tests, analyses, demonstrations, or inspections. Each verification activity references established engineering or regulatory standards and produces traceable evidence recorded in the System Requirements Verification Matrix (SRVM).

Verification Methods

- **T – Test:** Direct empirical measurement using calibrated instrumentation.
- **A – Analysis:** Mathematical, computational, or simulation-based evaluation.
- **D – Demonstration:** Operational proof of capability under representative conditions.
- **I – Inspection:** Visual or documentary verification of conformance to design or configuration data.

6.1 Performance Requirement

6.1.2 Swarm Deployment (A, D)
Network-layer modeling in MATLAB / ROS 2 simulation verifies mesh topology stability and latency < 1 s (Analysis). A live multi-unit demonstration confirms autonomous discovery and coordinated operation under open-water conditions (Demonstration).

6.1.3 Stability (T, A)
Buoyancy and self-righting capability validated per **ISO 12217-3:2022 Category C**. Analytical stability calculations (A) use CAD hydrostatics; physical tilt and capsize recovery tests (T) confirm upright recovery ≤ 10 s.

6.1.4 Speed (T)
Propulsion performance verified through on-water GPS trials. Speed vs. thrust logged at full load; acceptable range ≥ target ± 10 %. Conducted in calm Sea State ≤ 2.

6.1.5 Battery Life (T, A)
Continuous runtime testing at representative cruise load verifies endurance ≥ 5 min. Analytical validation (A) uses measured Wh consumption correlated to modeled energy budget from **POWER Subsystem Rev 1.0**.

6.1.6 Data Collection (T, I)
Bench and in-water testing confirm all sensors operate at ≥ 1 Hz sampling. Inspection checks calibration certificates per **ISO 22013:2021 Marine Sensor Performance**.

6.1.7 Data Mapping (D)
Demonstration on ground-station GUI confirms GPS/sensor data georeferenced within ≤ 5 m error and updated ≤ 5 s latency.

6.1.8 Obstacle Avoidance (D)
Pool demonstration verifies sensor detection ≥ 0.5 m ahead and automatic propulsion cutoff. Test filmed and logged for verification evidence.

6.2 Design Requirements

6.2.1 Hull Shape (T, A, I)
Hydrodynamic drag and self-righting analyzed in SolidWorks Flow Simulation (A). Scale-model towing tank tests (T) confirm coefficient-of-drag within ± 10 %. Inspection (I) verifies hull matches CAD and ISO 12215-2:2002 material specs.

6.2.2 Internal Volume (A, I)
CAD volume analysis (A) ensures internal layout supports all subsystems with ≥ 10 % buoyancy margin. Inspection (I) measures actual displacement and confirms fit within 0.07 m³ hull volume for main hull (with Starlink Antenna) and 0.001m³ for subunits (no Starlink Antenna).

6.2.3 Materials (I)
Material certifications checked for compliance with **ISO 12215-2:2002** and marine-grade corrosion standards. Visual inspection ensures absence of cracks, voids, or resin delamination.

6.2.4 Payload Mounts (T, I)
Static load test applies 1.5× payload mass for 60 s per **ASTM E8** guidance; no permanent deformation allowed. Inspection verifies mounting hardware torque and corrosion protection.

6.3 Environmental Requirements

6.3.1 Survivability (T)
Mechanical and environmental durability tested per **MIL-STD-810H**:

- Shock (Method 516.8 Proc IV)
- Vibration (Method 514.8 Cat 20 – Shipboard/Transport)
- Salt Fog (Method 509.7)

- Humidity (Method 507.6).

  All units must remain fully functional post-test.

6.3.2 Waterproofing (T, I)
Immersion test per **IEC 60529:2013 (IP68)**; 1.5 m depth for 30 min. No water ingress permitted. Inspection verifies gasket compression and connector seals.

6.3.3 Operating Temperature (T, A)
Thermal performance verified through **MIL-STD-810H Method 501.7 (High Temp)** and **502.7 (Low Temp)** chamber tests (0°C). Thermal finite-element analysis (A) validates internal heat dissipation.

6.3.4 Non-Hazardous Surface (I)
Inspection ensures exterior surfaces smooth, free of burrs, and compliant with handling-safety requirements per **ISO 13857** guidelines for edge radii.

6.4 Interface Requirements

6.4.1 Autonomy & Control (T, A)

Simulation (A) in ROS 2 verifies closed-loop stability and path tracking. Field test (T) confirms heading control within ± 5° and response latency ≤ 200 ms.

6.4.2 Inter-Drone Communication (T)

Functional test validates Wi-Fi (OpenWrt mesh) and optional LoRa mesh (Meshtastic) communication. Must maintain QoS 1 MQTT packets with ≤ 1 s latency over ≥ 200 m LOS. Logged via network sniffer.

6.4.3 Satellite Uplink (T, I)

End-to-end data transfer through **Starlink Ku-band** link confirms sustained ≥ 2 Mbps bandwidth and < 150 ms latency. Inspection ensures Ethernet/IP68 connectors per **IEC 60945** EMI sealing.

6.4.4 Display (D, I)

Demonstration of GUI verifies real-time telemetry, map overlays, and video feed. Inspection confirms all required parameters visible and labeled correctly.

6.4.5 Swarm Control (D, T)

Each drone shall publish and subscribe to swarm state messages at ≥ 1 Hz with < 1 s latency.

6.5 Verification Deliverables

Each verification activity will produce a controlled artifact stored in the project repository:

- **Test Reports:** procedure, instrumentation, data, photos, pass/fail summary.
- **Analysis Files:** CFD / thermal / stability models with assumptions and results.
- **Inspection Records:** checklists, material certifications, dimensional data.
- **Demonstration Evidence:** videos, telemetry logs, and operator sign-offs.

Completion of verification shall be achieved when every SR-ID entry in the SRVM is marked **PASS**, with supporting evidence reviewed by the subsystem leads and recorded under configuration control.

6.6 Verification Schedule and Responsibility Matrix

Verification tasks will be executed in accordance with the project's development timeline (prototype build, bench testing, field validation, and documentation review).

Each activity is assigned to a responsible member to ensure traceable ownership, repeatability, and documentation of results.

All findings will be entered into the SRVM upon completion.

Verification ID / Requirement Category	Verification Method(s)	Primary Responsibility	Planned Phase	Deliverable
5.1 Performance Requirements				
5.1.2 Swarm Deployment	Analysis / Demonstration	John Lopez (Software & Comms)	Field Phase II – Swarm Trial	Network Latency Log + Demo Video
5.1.3 Stability	Test / Analysis (ISO 12217-3)	Kama Henriques(Mechanical Design)	Prototype Phase I – Pool Testing	Stability Curve Plot + Righting Video
5.1.4 Speed	Test (GPS Trials)	Robert Wilkie(Propulsion)	Field Phase II – Sea Trial	Speed vs Thrust Report
5.1.5 Battery Life	Test / Analysis	John Lopez(Power Systems)	Bench Phase I – Endurance Run	Energy Budget Validation Sheet
5.1.6 Data Collection	Test / Inspection (ISO 22013)	Jamie Ledbetter (Sensors Lead)	Bench Phase I – Calibration	Sensor Calibration Records
5.1.7 Data Mapping	Demonstration	Mayra Mendoza (Software & Comms)	Integration Phase III	GUI Mapping Test Video
5.1.8 Obstacle Avoidance	Demonstration	Jamie Ledbetter (Mechanical Design)	Field Phase II	Obstacle Test Log and Video
5.2 Design Requirements (All)	Analysis / Inspection / Test	Jamie Ledbetter + Mayra Mendoza	Design Phase 0 – Review and Prototyping	Design Compliance Checklist
5.3 Environmental Requirements (All)	Test / Analysis / Inspection (MIL-STD-810H ; IEC 60529)	John Lopez (Power) + Robert Wilkie(Propulsion)	Qualification Phase III	Environmental Test Report #ENV-01

Verification ID / Requirement Category	Verification Method(s)	Primary Responsibility	Planned Phase	Deliverable
5.4 Interface Requirements (5.4.1 & 5.4.2)	Test / Demonstration / Inspection	Mayra Mendoza (Comms) + Jamie Ledbetter (Sensors)	Integration Phase III	Interface Validation Summary
5.4.3 Satellite Uplink	Test / Inspection (IEC 60945)	John Lopez (Comms)	Field Phase II	Bandwidth and EMI Compliance Record
5.4.4 Display	Demonstration / Inspection	Mayra Mendoza (Comms & GUI)	Integration Phase III	GUI Verification Form

Execution Timeline Summary

1. **Design Phase 0 (Weeks 1–5):** Analytical verifications (A, I); CAD, materials, layout, safety checks.
2. **Prototype Phase I (Weeks 6–12):** Environmental and component testing (T); waterproofing, drop, power endurance.
3. **Field Phase II (Weeks 13–18):** Operational demonstrations (D); swarm networking, stability, obstacle avoidance, uplink.
4. **Integration Phase III (Weeks 19–24):** Final interface and GUI verification; compile SRVM evidence and verification reports.

Completion Criteria:
All verification tasks are considered complete when:

- Data meets defined pass/fail thresholds;
- Reports are reviewed by at least one independent member;
- Results and evidence are archived under configuration control;
- SRVM entries are updated to “PASS.”

7.0 System Requirements Verification Matrix (SRVM)

Requirement ID / Description	Verification Method T = Test, A = Analysis, D = Demonstration, I = Inspection			
4.1 Performance Requirements	**T**	**A**	**D**	**I**
4.1.2 Swarm Deployment			X	
4.1.3 Stability	X		X	
4.1.4 Speed	X			
4.1.5 Battery Life	X	X		
4.1.6 Data Collection	X		X	
4.1.7 Data Mapping		X		
4.1.8 Obstacle Avoidance	X		X	
4.2 Design Requirements	**T**	**A**	**D**	**I**
4.2.1 Hull Shape		X		
4.2.2 Internal Volume		X		X
4.2.3 Materials				X
4.2.4 Payload Mounts			X	X
4.3 Environmental Requirement	**T**	**A**	**D**	**I**
4.3.1 Survivability	X	X		
4.3.2 Waterproofing	X			X
4.3.3 Operating temperature	X	X		
4.3.4 Non-Hazardous Surface				X
4.4 Interface Requirements	**T**	**A**	**D**	**I**
4.4.1 Autonomy & Control			X	
4.4.2 Inter-Drone Communication	X		X	
4.4.4 Satellite Uplink	X		X	
4.4.5 Display			X	X

8.0 Constraints

8.1 Budget:

Prototype cost limited to approximately $4,000. Total power budget sized for ≥ 250 W continuous and ≥ 600–800 W transient capacity to remain within component and thermal limits.

8.2 Mass / Buoyancy:

Displacement ≈ 0.07 m^3 → ≈ 71 kg buoyancy in seawater. Target fully equipped mass ≤ 24 kg using $LiFePO_4$ battery chemistry to maintain positive stability margin.

8.3 Safety:

All critical functions must fail-safely. A local hardware E-stop shall override propulsion and power outputs independent of the cloud link. No single-point failure shall result in loss of control or uncontained energy release.

8.4 Regulatory:

- Operation in navigable waters must follow **COLREGs** lighting and right-of-way requirements for unmanned surface craft.
- AIS transmissions, if used, must conform to **ITU-R M.1371 / IEC 61993-2** and be licensed under national radio regulations (e.g., FCC Part 80).
- Electrical enclosures must meet **IEC 60529 (IP68)** ingress-protection standards.
- All testing and documentation shall adhere to **MIL-STD-810H** environmental methods where applicable.

8.5 Environmental:

- Materials and deployment practices must comply with **IMO MARPOL Annex V**, prohibiting the release of plastics or non-biodegradable debris.
- The system must withstand operation in Sea State ≤ 3, ambient –10 °C to 50 °C, and salt-fog exposure per **MIL-STD-810H Method 509**.
- Battery chemistry and waste disposal must conform to **UN/DOT 38.3** transport safety and local hazardous-waste handling rules.

8.6 Sponsor / Programmatic:

- All documentation, testing, and deliverables must align with **University of Arizona ENGR 498A/B** capstone requirements and grading rubric.
- The final prototype must demonstrate minimum viable functionality (self-righting, data acquisition, uplink) within the semester schedule.
- Use of components must remain within sponsor-approved budget and comply with procurement policies for COTS hardware.

8.7 Operational:

- Communication links shall use approved frequencies and encryption (MQTT over TLS 1.2).
- Human operators must maintain remote manual override at all times during testing.

9.0 Notes and Assumptions

The following notes and assumptions establish boundary conditions, dependencies, and expectations used during system definition and verification.

9.1 Operational Assumptions

1. The AQUABOT will operate primarily at the air–water interface; brief submersion (< 5 s) may occur due to waves or impact but full underwater navigation is not required.
2. Sea State ≤ 3 (wave height ≤ 1.25 m) and sustained wind ≤ 15 knots during normal operations.
3. The system will be operated only in approved test or mission areas cleared with aviation and maritime authorities.
4. Starlink satellite coverage and sufficient network connectivity are assumed to be available at deployment latitude.
5. Mission duration limited to ≈ 5 h endurance; PV charging used only for life-extension and not relied upon for continuous power.

9.2 Design and Verification Assumptions

1. The hull's center of gravity and center of buoyancy will remain within ±5 % of modeled design after component installation.
2. All COTS electronic components will perform within manufacturer-rated temperature and humidity limits.
3. Calibration drift of sensors during a single mission (< 6 h) is negligible and will not materially affect data accuracy.
4. Environmental verification (MIL-STD-810H) will be conducted on a single prototype representing the production configuration.
5. Laboratory pool and small-lake testing are assumed to be representative of full-scale sea operation for verification purposes.
6. All verification instrumentation (load cells, IMUs, GPS loggers) will be traceable to NIST or equivalent calibration.

9.3 Regulatory and Programmatic Notes

1. AIS activation will only occur under authorized frequency licenses; otherwise a simulation mode will be used for testing.
2. LoRa mesh radios will operate within unlicensed ISM bands (< 1 W EIRP) and follow FCC Part 15 limits.

3. All project results, data, and intellectual property remain under the University of Arizona ENGR 498 Capstone Program per sponsor agreement.
4. All safety documentation, test reports, and SRVM evidence will be retained for at least two academic years for audit or reuse.
5. Any future integration with military or agency programs may require additional certification (MIL-STD-882E safety program, STANAG 4748 acoustic link).

Appendix A – Subsystem Summaries

Subsystem	Primary Functions / Highlights
Propulsion	Dual brushless thrusters, ESC control via ESP32 PWM; designed for ≤ 40 A pk each.
Flight Controller (ESP32)	IMU-based stabilization, GPS interface, E-stop, self-righting logic, PWM output.
Sensors & Sampling	pH, EC, temperature, humidity, sonar; integrated through I²C/UART @ 3.3 V.
Mission Computer (Jetson Nano)	ROS 2 Foxy nodes for data fusion, telemetry, and video encoding (RTSP/MJPEG).
Satellite Uplink (Starlink)	30–40 W avg load; Ethernet interface; primary telemetry path to cloud broker.
Power Subsystem	$LiFePO_4$ battery (0.8–1.3 kWh) + PV offset (50–120 W); regulated 5 V rails.
Self-Righting & Failsafes	Automatic recovery using mass distribution and thrust vectoring; local E-stop.
Ground Station & Cloud	MQTT broker with GUI dashboard, video pane, and heatmap visualization.

Appendix B – Power Budget and Energy Distribution

Mode	Propulsion (W)	House Loads (W)	Total Power (W)	5 h Energy (Wh)	Notes
Idle / Loiter (sunny +PV)	30	58	≈ 38 net	190	PV offset ~50 W
Survey Cruise (rec.)	100–150	58	158–208	790–1040	Nominal mission mode
Worst-Case Transient	300	70	≈ 370	N/A	< 10 s bursts

Appendix C – Applicable Standards and Regulations Summary

Standard / Regulation	Short Description / Scope
IEC 60529:2013	IP Code for enclosure waterproofing (IP68 target).
MIL-STD-810H (2019)	Environmental tests: shock, vibration, temperature, humidity, salt fog.
MIL-STD-461G (2015)	EMI/EMC control for electronic equipment.
ASTM F3322-18	sUAS parachute design and verification tests.
IMO MARPOL Annex V	Ban on plastic and garbage discharge at sea.
ITU-R M.1371	AIS technical characteristics (VHF protocols).iso 1221
IEC 61993-2	AIS Class A equipment performance standard.
ISO 12215-2:2002	Hull materials and sandwich core requirements.
ISO 12217-3:2022	Small craft stability and buoyancy assessment.
ASTM F3052-14(2020)e1	Small boat stability (air-inclining experiment).
ISO 22013:2021	Marine sensor performance and testing.
COLREGs (1972)	Maritime collision-avoidance rules (lights / right-of-way).
UN/DOT 38.3	Lithium battery transport safety testing.
IEC 60945	Maritime electronics environmental and EMC testing.
MIL-STD-882E	System safety program and hazard analysis process.
MIL-STD-464C	System-level electromagnetic environment effects.